

Jason Hoang

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Technical Skills

- **Languages:** Python, C, C++, C#, SQL, HTML, CSS, JavaScript
- **Databases:** MongoDB, PostgreSQL
- **Framework/Tools:** Unity, Git, Jira, Jenkins, Confluence, Linux, VS Code, NumPy, Pandas, Agile/Scrum, React

Experience

Junior Software Engineer – MDA Space

July 2022 – September 2022

- Identified, reproduced, and tracked software bugs in Jira, with detailed documentation in Confluence for development teams
- Participated in daily scrum meetings, providing updates on software development, testing progress and bug statuses
- Developed and deployed real-time flight warning systems in C++, detecting obstacles within 5,000 feet with 100% accuracy
- Refactored 3,000+ lines of legacy C++ code, reducing memory usage, and improving execution time

Projects

Tetris Arcade Game – Unity, C#

March 2025

- Designed and executed comprehensive test plans, including functional, regression, and manual playtesting, ensuring a high-quality game experience
- Identified, documented, and tracked 10+ unique bugs using Jira and Microsoft Word
- Implemented core game mechanics, including movement, rotation, gravity, line clearing, scoring, and hard drops
- Utilized Git for version control to store project software

Pacman Arcade Game – Unity, C#

January 2025

- Developed and executed test plans for functional, regression, and manual playtesting
- Identified, documented, and tracked 15+ unique bugs using Jira and Microsoft Word
- Developed AI-driven ghost behaviours (chase, scatter, flee) for an interactive game environment
- Integrated 2D animations and physics-based collision detection for smooth gameplay
- Implemented complete audio systems, including music, sound effects and dynamic feedback

Minesweeper Puzzle Game – Depth First Search – Unity, C#

September 2024

- Developed a scalable Minesweeper Game with dynamic board resizing, randomized mine placement, and interactive user input handling
- Implemented a depth-first search (DFS) flood fill algorithm, effectively revealing adjacent empty tiles in $O(n*m)$ time complexity, improving game responsiveness

Tic Tac Toe Game – Monte Carlo Tree Search – Python

June 2024

- Implemented Monte Carlo Tree Search AI in Python, simulating 5,000+ games per turn to achieve 99.9% win rate, demonstrating advanced search algorithms used in real-world decision-making AI
- Conducted manual playtesting and documented data in Microsoft Excel

Education

Simon Fraser University - Bachelor of Science, Major in Computing Science

August 2021